

(REVISION-2021)

Reg.No

Signature

MODEL QUESTION PAPER (REV-21)

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

PROBLEM SOLVING USING PYTHON-SET-I

(Maximum Marks : 75)

TIME : 2HRS

PART -A

Answer **all** questions in one word or one sentence.

I.

1. List any two standard datatypes in python.
2. List the symbols used in drawing flowchart.
3. Define recursive function.
4. Do loop statements have else clause? When will it be executed?
5. Write any two methods that are used in Python Lists.
6. What is the output of print tuple + tinytuple if tuple = ('abcd', 786 , 2.23, 'john', 70.2) and tinytuple = (123, 'john')?
7. Write methods to rename and delete files.
8. What is a pickle module in python?
9. Distinguish between local variable and global variable.

(1*9=9marks)

PART B

Answer any **eight** questions from the following. Each question carries 3 marks.

II.

1. Mention the features of identity operators?
2. State how to perform comments in Python.
3. What is len function and explain how it is used on strings with an example .
4. Explain about string slicing with examples.
5. How does del operation work on dictionaries? Give an example.
6. What are the advantages of Tuple over List?
7. Explain the concept of builtin exceptions.
8. How can you set a file offset?
9. Explain with an example how strings are used in python?
10. List any three rules for declaring variables in python.

I

(8*3=24)

PART C

Answer **ALL** questions. Each question carries 7 marks

III. Mention the different types of iterative structure allowed in python. Explain the use of continue and break statements with an example.

OR

IV. What is an algorithm. Summarize the characteristics of a good algorithm. Outline the algorithm for displaying the first n even numbers.

V. Find the area and perimeter of a circle using functions. Prompt the user for input.

OR

VI. Explain type conversion and type coercion in python.

VII. Write a python program to find the size of a tuple.

OR

VIII. What is Dictionary? Explain Python dictionaries in detail discussing its operations and methods.

IX. What are exceptions? Explain the method to handle them with examples.

OR

X . Write a python code to count the number of words in a python file.

XI. Analyse string slicing. Illustrate how it is done in python with example.

OR

XII. Outline about function definition and call with example. Why are functions needed?

XIII. Summarize the precedence of mathematical operators in python.

OR

XIV. Draw a flowchart to accept three distinct numbers, find the greatest and print the result.

(6*7=42)

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MODEL QUESTION PAPER (REV-21)

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

PROBLEM SOLVING USING PYTHON-SET-II

(Maximum Marks : 75)

TIME : 2HRS

PART -A

Answer **all** questions in one word or one sentence.

1. Give the syntax for if statement in Python.
2. Give the output for the following expression evaluation in Python:
(i) $-4^{**}2$ (ii) $2^{**}3^{**}2$
3. What is recursion?
4. What are default arguments?
5. Discuss type coercion
6. Explain any 2 Dictionary methods.
7. Explain the use of functions len() and max() in list.
8. What is pickling?
9. How can we open and close files in Python?

(1*9=9marks)

PART B

Answer any **eight** questions from the following. Each question carries 3 marks.

II.

1. Differentiate between break and continue statements with proper examples.
2. Explain various datatypes in Python.
3. Write a Python program to find the factorial of a given number.
4. Why do we need functions? What are the advantages of using a function?
5. Explain how dictionaries are created.
6. Write a Python code to search an element in a list
7. Write the syntax for opening a file in Python. Give one example.
8. Explain built-in exceptions.
9. List the advantage of using pickling in python. Also state the usage of dump method with suitable examples.
10. What are the key features of python?

(8*3=24)

PART C

Answer **ALL** questions. Each question carries 7 marks

III. . Draw the flow chart for finding largest of three numbers and write an algorithm

OR

IV. Discuss different operators used in Python. Differentiate precedence and associativity of operators.

V. Explain the working of while loop with example. Write a program to find the sum of digits of a number using while loop.

OR

VI. Write a menu driven program to calculate the area of circle, rectangle and square. Use a separate function to implement each operation.

VII. List out different string operators in Python.

OR

VIII. Write Python program to count the total number of vowels, consonants and blanks in a String.

IX. Explain any 5 built-in dictionary functions with examples.

OR

X. Write a Python program to create a dictionary of roll numbers and names of 5 students.

Display the contents of dictionary in alphabetical order of names.

XI. What is tuple in Python ? How does it differ from List?

OR

XII. With examples explain the built-in methods used for list operations.

XIII. Write a Python program to create a text file. Read the contents of the file, encrypt every character in the file with a distance of 3 and write it to a new file.

OR

XIV. Explain how exceptions are handled in Python with illustrations.

(6*7=42)